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OS Practice Week 10

1. Create a hidden configuration file named .net_config in your home directory, add a sample IP address inside, and verify it appears only when listing all files.

```
nurarys@nurarys-VirtualBox:~$ echo "IP=192.168.1.23" > .netconfig
nurarys@nurarys-VirtualBox:~$ ls -a | grep .netconfig
.netconfig
```

2.Display the full path, type, and permissions of the /dev/null, /dev/random, and /dev/zero devices, and explain their purposes briefly.

```
nurarys@nurarys-VirtualBox:~$ ls -l /dev/null /dev/rendom /dev/zero
ls: cannot access '/dev/rendom': No such file or directory
crw-rw-rw- 1 root root 1, 3 Nov 11 08:11 /dev/null
crw-rw-rw- 1 root root 1, 5 Nov 11 08:11 /dev/zero
nurarys@nurarys-VirtualBox:~$ file /dev/null /dev/rendom /dev/zero
/dev/null:   character special (1/3)
/dev/rendom: cannot open `/dev/rendom' (No such file or directory)
/dev/zero:   character special (1/5)
```

3.Create a file /home/user1/network_info.txt and add the output of ip addr show into it in one command.

```
nurarys@nurarys-VirtualBox:~$ ip addr show | sudo tee /home/user1/network_info.txt >/dev/null
nurarys@nurarys-VirtualBox:~$ cd /home/user1
nurarys@nurarys-VirtualBox:/home/user1$ cat network_info.txt
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default q
len 1000
    link/ether 08:00:27:c3:00:ee brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85669sec preferred_lft 85669sec
    inet6 fd17:625c:f037:2:157b:eb16:c883:817d/64 scope global temporary dynamic
       valid_lft 86183sec preferred_lft 14183sec
    inet6 fd17:625c:f037:2:a00:27ff:fec3:ee/64 scope global dynamic mngtmpaddr
       valid_lft 86183sec preferred_lft 14183sec
    inet6 fe80::a00:27ff:fec3:ee/64 scope link
       valid_lft forever preferred_lft forever
```

4.Create symbolic and hard links to /etc/hosts named hosts_sym and hosts_hard. Then edit /etc/hosts and check which links reflect the changes.

```
nurarys@nurarys-VirtualBox:~$ ln -s /etc/hosts  ~/hosts_SYM
nurarys@nurarys-VirtualBox:~$ sudo ln /etc/hosts ~/hosts_HARD
nurarys@nurarys-VirtualBox:~$ echo '# lab test' | sudo tee -a /etc/hosts >/dev/null
nurarys@nurarys-VirtualBox:~$ echo '#--- SYM ---'; head -n 3 ~/hosts_SYM
#--- SYM ---
127.0.0.1 localhost
127.0.1.1 nurarys-VirtualBox

nurarys@nurarys-VirtualBox:~$ echo '#--- HARD ---'; head -n 3 ~/hosts_HARD
#--- HARD ---
127.0.0.1 localhost
127.0.1.1 nurarys-VirtualBox
```

5.Display all active network connections with process names and listening ports, and save the output into net_connections.txt.

```
nurarys@nurarys-VirtualBox:~$ sudo ss -tunap > ~/net_connections.txt
nurarys@nurarys-VirtualBox:~$ cat ~/net_connections.txt
Netid State  Recv-Q Send-Q    Local Address:Port  Peer Address:PortProcess

udp   UNCONN 0      0               0.0.0.0:5353       0.0.0.0:*    users:(("avahi-daemon",pi
d=721,fd=12))
udp   UNCONN 0      0            127.0.0.54:53         0.0.0.0:*    users:(("systemd-resolve"
,pid=381,fd=16))
udp   UNCONN 0      0         127.0.0.53%lo:53         0.0.0.0:*    users:(("systemd-resolve"
,pid=381,fd=14))
udp   ESTAB  0      0      10.0.2.15%enp0s3:68        10.0.2.2:67   users:(("NetworkManager",
pid=807,fd=26))
udp   UNCONN 0      0               0.0.0.0:37967      0.0.0.0:*    users:(("avahi-daemon",pi
```

6.Create a simple HTTP server in your home directory using Python on port 8081, access it via curl, and save the result to server_test.log.

```
nurarys@nurarys-VirtualBox:~$ (cd ~ && python3 -m http.server 8081 >/dev/null 2>&1 & echo $! > ~/http8081.pid)
nurarys@nurarys-VirtualBox:~$ curl -s http://127.0.0.0:8081/ > ~/server_test.log
nurarys@nurarys-VirtualBox:~$ cat ~/server_test.log
<!DOCTYPE HTML>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href=".bash_history">.bash_history</a></li>
<li><a href=".bash_logout">.bash_logout</a></li>
```

7.Explore the /proc/net directory and list all files containing the word “tcp” in their name. Redirect the output to tcp_files.txt.

```
nurarys@nurarys-VirtualBox:~$ cat ~/tcp_files.txt
tcp
tcp6
```

8. Create a script net_backup.sh that archives /etc/network into /home/$USER/net_backup.tar.gz, prints “Backup Complete”, and make it executable.

```
nurarys@nurarys-VirtualBox:~$ nano net_backup.sh
nurarys@nurarys-VirtualBox: ~
  GNU nano 7.2                    net_backuo.sh *
#!/bin/bash
tar -czf /home/$USER/net_backup.tar.gz /etc/network 2>/dev/null
echo "Backup Complate"
nurarys@nurarys-VirtualBox:~$ sudo ./net_backup.sh
Backup Complate
```

9.Set the sticky bit on /home/shared_network so that users can create and edit files but only delete their own. Verify it with ls -ld.

```
nurarys@nurarys-VirtualBox:~$ sudo mkdir -p /home/shared_network
nurarys@nurarys-VirtualBox:~$ sudo chmod 1777 /home/shared_network
nurarys@nurarys-VirtualBox:~$ ls -ld /home/shared_network
drwxrwxrwt 2 root root 4096 Nov 11 08:59 /home/shared_network
```

10.Write one command that checks if port 22 (SSH) is open on localhost using ss or netstat, and append the result (open/closed) into ssh_status.log.

```
nurarys@nurarys-VirtualBox:~$ ( sudo ss -tln | grep -qE 'LISTEN.*:22(\s|$)' && echo "$(date) open"
 || echo "$(closed)" ) >> ~/ssh_status.log
nurarys@nurarys-VirtualBox:~$ cat  ~/ssh_status.log
Tue Nov 11 09:04:27 AM +05 2025 open
```